Complete WSL + Docker + Datadog + Flask APM Setup Guide

Overview

This document contains:

- WSL installation and verification
- Ubuntu installation in WSL
- Docker Desktop installation
- Docker commands
- Python installation
- Flask application setup
- Datadog Agent installation in Docker
- Datadog APM tracing setup
- RUM setup
- Important Linux/Ubuntu commands
- Troubleshooting steps

This can be used as your future reference document.

---

1. CHECK WHETHER WSL IS INSTALLED

Open Windows PowerShell:

```
wsl --status
```

If WSL is installed, you will see:

```
Default Distribution: Ubuntu
Default Version: 2
```

---

2. INSTALL WSL + UBUNTU

Install WSL

Open PowerShell as Administrator:

```
wsl --install
```

Restart system after installation.

Install Ubuntu

Open Microsoft Store.

Search:

```
Ubuntu
```

Install latest Ubuntu version.

After installation:

Open Ubuntu.

Create:

- username
- password

---

3. IMPORTANT WSL COMMANDS

Check WSL distributions

```
wsl -l -v
```

Shutdown WSL

```
wsl --shutdown
```

Open Ubuntu

```
wsl
```

OR open Ubuntu app.

Check Linux version

```
uname -a
```

---

4. IMPORTANT UBUNTU COMMANDS

Current directory

```
pwd
```

List files

```
ls
```

Detailed list

```
ls -lrt
```

Change directory

```
cd folder_name
```

Go home directory

```
cd ~
```

Create folder

```
mkdir folder_name
```

Remove file

```
rm file_name
```

Remove folder

```
rm -rf folder_name
```

Open file editor

```
nano app.py
```

Save in nano

```
CTRL + O
```

Exit nano

```
CTRL + X
```

---

5. INSTALL DOCKER DESKTOP

Download:

https://www.docker.com/products/docker-desktop/

Install Docker Desktop.

Enable:

```
Use WSL 2 instead of Hyper-V
```

Restart system.

---

6. VERIFY DOCKER INSTALLATION

Open Ubuntu terminal:

```
docker --version
```

Check running containers

```
docker ps
```

Check all containers

```
docker ps -a
```

---

7. IMPORTANT DOCKER COMMANDS

Pull image

```
docker pull redis
```

Run container

```
docker run -d --name redis redis
```

Stop container

```
docker stop redis
```

Start container

```
docker start redis
```

Remove container

```
docker rm -f redis
```

Container logs

```
docker logs redis
```

Enter container

```
docker exec -it redis bash
```

---

8. INSTALL PYTHON IN UBUNTU

Update packages

```
sudo apt update
```

Install Python

```
sudo apt install python3 -y
```

Verify Python

```
python3 --version
```

Install pip

```
sudo apt install python3-pip -y
```

Verify pip

```
pip3 --version
```

---

9. CREATE PYTHON VIRTUAL ENVIRONMENT

Install venv

```
sudo apt install python3-venv -y
```

Create project folder

```
mkdir flask-apm-demo
cd flask-apm-demo
```

Create virtual environment

```
python3 -m venv venv
```

Activate environment

```
source venv/bin/activate
```

You will see:

```
(venv)
```

---

10. INSTALL FLASK + DATRADOG LIBRARIES

Install Flask

```
pip install flask
```

Install Datadog tracing

```
pip install ddtrace
```

Install flask-cors

```
pip install flask-cors
```

---

11. CREATE FLASK APPLICATION

Create file:

```
nano app.py
```

Paste:

```
from flask import Flask, jsonify
from flask_cors import CORS
import time
import random

app = Flask(__name__)
CORS(app)

@app.route('/')
def home():
    return jsonify({"message": "Hello AIOps!", "status": "ok"})

@app.route('/api/users')
def get_users():
    time.sleep(random.uniform(0.1, 0.5))
    return jsonify({"users": ["user1", "user2", "user3"]})
```

```python
@app.route('/api/orders')
def get_orders():
    time.sleep(random.uniform(0.3, 1.0))
    return jsonify({"orders": [101, 102, 103]})

@app.route('/api/slow')
def slow_endpoint():
    time.sleep(random.uniform(2.0, 4.0))
    return jsonify({"message": "slow response"})

@app.route('/api/error')
def error_endpoint():
    raise Exception("Simulated error for APM demo!")

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000, debug=False)
```

Save:

```
CTRL + O
ENTER
CTRL + X
```

---

12. INSTALL DATADOG AGENT IN DOCKER

Remove old container

```
docker rm -f datadog-agent
```

Run Datadog Agent

```
docker run -d
--name datadog-agent
-e DD_API_KEY=YOUR_API_KEY
-e DD_SITE="datadoghq.com"
-e DD_APM_ENABLED=true
-e DD_LOGS_ENABLED=true
-e DD_PROCESS_AGENT_ENABLED=true
-p 8126:8126/tcp
-v /var/run/docker.sock:/var/run/docker.sock
-v /proc/:/host/proc/:ro
```

```
-v /sys/fs/cgroup/:/host/sys/fs/cgroup:ro
datadog/agent:latest
```

Replace:

```
YOUR_API_KEY
```

with your Datadog API key.

---

13. VERIFY DATADOG AGENT

Check container

```
docker ps
```

Verify APM port

```
curl http://192.168.64.1:8126/info
```

If successful, JSON output appears.

---

14. START FLASK APP WITH DATRADOG APM

Go to project

```
cd ~/flask-apm-demo
```

Activate venv

```
source venv/bin/activate
```

Export environment variables

```
export DD_SERVICE=flask-perf-demo
export DD_ENV=dev
export DD_VERSION=1.0
export DD_AGENT_HOST=192.168.64.1
export DD_TRACE_AGENT_PORT=8126
```

Start app with tracing

```
ddtrace-run python3 app.py
```

---

15. TEST FLASK APIs

Home API

```
http://127.0.0.1:5000/
```

Users API

```
http://127.0.0.1:5000/api/users
```

Slow API

```
http://127.0.0.1:5000/api/slow
```

Error API

```
http://127.0.0.1:5000/api/error
```

---

16. CHECK APM IN DATADOG

Go to:

```
Datadog → APM → Services
```

Service:

```
flask-perf-demo
```

---

17. PYTHON INSTALLATION IN WINDOWS

Install Python

Download:

https://www.python.org/downloads/windows/

IMPORTANT:

Enable:

```
Add Python to PATH
```

Verify Python

Open PowerShell:

```
python --version
```

OR

```
py --version
```

---

18. START PYTHON WEB SERVER FOR RUM HTML

Go to Downloads:

```
cd C:\Users\LENOVO\Downloads
```

Start server

```
py -m http.server 8080
```

Open browser

```
http://localhost:8080/rum-apm-dashboard.html
```

---

19. DATADOG RUM SETUP

In HTML add:

```
clientToken
applicationId
```

from:

```
Datadog → UX Monitoring → Applications
```

---

20. IMPORTANT TROUBLESHOOTING COMMANDS

Check port usage

```
lsof -i :5000
```

Kill process

```
kill -9 PID
```

Check Docker logs

```
docker logs datadog-agent
```

Restart WSL

```
wsl --shutdown
```

Restart Docker Desktop

Close and reopen Docker Desktop.

---

21. COMPLETE ARCHITECTURE

```
Browser (RUM)
       ↓
Frontend HTML Dashboard
       ↓
Flask APIs (WSL Ubuntu)
       ↓
Datadog Agent (Docker)
       ↓
Datadog Cloud
```

---

22. WHAT YOU LEARNED

Infrastructure Monitoring

- CPU
- Memory
- Processes
- Docker metrics

APM

- Traces
- Spans
- API latency

- Errors

RUM

- User sessions
- Clicks
- Browser monitoring
- Frontend performance

Troubleshooting

- WSL networking
- Docker networking
- CORS issues
- Port conflicts
- Agent connectivity

---

23. FUTURE LEARNING TOPICS

- Datadog Dashboards
- SLO/SLA Monitoring
- Synthetic Monitoring
- Kubernetes Monitoring
- Redis Monitoring
- Database Monitoring
- CI/CD Integration
- Jenkins + JMeter
- Grafana
- Dynatrace
- Splunk
- OpenTelemetry

---

24. STOP SERVICES

Stop Flask App

```
CTRL + C
```

Stop Datadog Agent

```
docker stop datadog-agent
```

Stop Python HTTP Server

```
CTRL + C
```

---

25. START SERVICES AGAIN

Start Datadog Agent

```
docker start datadog-agent
```

Start Flask App

```
cd ~/flask-apm-demo
source venv/bin/activate

export DD_SERVICE=flask-perf-demo
export DD_ENV=dev
export DD_VERSION=1.0
export DD_AGENT_HOST=192.168.64.1
export DD_TRACE_AGENT_PORT=8126

ddtrace-run python3 app.py
```

Start HTTP Server

```
cd C:\Users\LENOVO\Downloads
py -m http.server 8080
```

END